

Compositionality: hierarchy, and the pressures for breadth and depth

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Draft of July 29, 2026

1 What is compositionality?

There is a generally shared intuition that the ability to compose outputs, behaviours or functions is a powerful technique to create more capable systems, and there has been a fair amount of work studying how compositional LLMs are at a behavioral level. However, there is a puzzle in applying this lens to the internal circuitry of a neural network. A deep network is, by definition, a composition of functions and so automatically compositional under our above definition.

However, there is a clear difference between different networks in how compositional they seem to our intuition (as we will see in examples below). I will argue that what we really want (as suggested by Danaia) is that the network solves a task family by combining a few **computational units** (i.e., functions or subcircuits part of the computational graph of the trained network), each useful across many tasks — so that when a new task arrives, the expressivity *reachable in a short period of training* is much higher than raw parameter count suggests. This ties the decomposition of the circuit to the structure of the task, and therefore makes the decomposition not automatic.

Before going further we fix the basic objects.

Definition 1 (network, parameters). A neural architecture defines a differentiable map $f(x; \theta)$ from inputs to output logits, where $\theta \in \mathbb{R}^P$ is the concatenation of all trainable weights. Training is stochastic gradient descent on $L(\theta) = \mathbb{E}_x \ell(x; \theta)$ for a per-example loss ℓ (cross-entropy for the sequence tasks; the B2 experiment below is a regression and uses squared loss). Whenever we write a gradient below, it is a vector in $\mathbb{R}^P$, taken with respect to this θ at the current training point θ_t .

Definition 2 (circuit). The computation of $f(\cdot; \theta)$ unrolls into a directed acyclic computational graph whose nodes are intermediate quantities (neurons, attention scores, residual-stream coordinates at a given layer and position) and whose edges carry the weight entries that connect them. A *circuit* C is a subgraph together with the designation of its edge weights as free: the circuit’s tangent space is the coordinate subspace of $\mathbb{R}^P$ spanned by C ’s free weights, and P_C denotes the coordinate projection onto it (so $P_C g$ is simply the gradient restricted to C ’s weights). In practice we avoid attributing individual weights to circuits by working with *interface circuits*: C is specified by a site (layer, position class) and a feature map ϕ , its free parameters being a virtual zero-initialized adapter at that site (Definition 7); the two views agree when the subgraph is the one feeding the interface.

Definition 3 (breadth and depth). The **breadth** of a circuit is the number of distinct computational units that feed into it, and its **depth** is the number of computational units it serially consumes before reaching the input.

Examples will make clear how we are using these terms; in this draft they are organizing vocabulary rather than measured quantities. We first explain a possible incentive for neural networks to learn compositionally broad circuits, coming directly from the number of distinct ‘tasks’ the network needs to solve.

*Draft; experiments and analysis joint with Claude. Code: github.com/asving/RRXOR, [correction_geometry/](https://github.com/asving/correction_geometry/).

2 Breadth: the pressure to share across tasks

The simplest setting is when each circuit is the solution to a particular task in our task family. For instance, in a next-token prediction task on a sequence generated by an HMM, the tasks might be indexed by the hidden state of the system (or some coarsening of such). The network then has to learn a different solution conditional on the hidden state, learn to estimate its belief distribution over these hidden states, and use this distribution to combine the different solutions. Let us first investigate the potential learning pressures in this simple compositional circuit.

Example 1 (RRXOR). An example to keep in mind throughout is the RRXOR sequence, of the form $\dots \{\text{random bit}\} \{\text{random bit}\} \{\text{xor of the previous two}\} \dots$. Precisely: draw iid fair bits $r_1^{(k)}, r_2^{(k)}$ and form blocks $B^{(k)} = (r_1^{(k)}, r_2^{(k)}, r_1^{(k)} \oplus r_2^{(k)})$; concatenate the blocks and emit a window of length L starting at a uniformly random offset $o \in \{0, 1, 2\}$. Each position t then carries a hidden phase $p_t = (o + t) \bmod 3$: positions with $p_t \in \{0, 1\}$ are free bits, and $x_t = x_{t-1} \oplus x_{t-2}$ whenever $p_t = 2$. One way to classify the tasks is as ‘predicting a random bit’ and ‘predicting the xor of the previous two bits’; which prediction applies is indexed by a belief over the current phase. The generator’s minimal causal machine has 5 states (nothing pending / one pending bit $\times 2$ / pending xor $\times 2$). An observer ignorant of o must synchronize: its belief state is the posterior over phase together with the last two bits, and exact enumeration of this filter gives 51 reachable states which minimize to exactly **36** under bisimulation, i.e. keeping only information relevant for predicting the future. The Bayes floor is $\ln 2$ at free positions and 0 at xor positions once synchronized.

Each ‘solution’ receives positive gradient pressure on the tokens corresponding to its hidden states. On other hidden states, we might simplify the analysis by assuming that the gradient pressure is a random vector, and expect that the sum of the gradient contribution over all the other hidden states cancels out relative to the positive gradient pressure.

From this perspective, any circuit in the network receives a mean gradient relative to each task, and the total amplification it receives depends on the total gradient vector across all tasks. This will be large if there are many tasks with an ‘aligned’ mean-gradient vector, i.e., the way the circuit is used across different tasks is roughly identical. On the contrary, the total contribution will be ≈ 0 if the circuit is used in a different way in each task.

Definition 4 (gradient vectors). For a task τ with data distribution D_τ and per-example loss $\ell_\tau(x; \theta)$, distinguish three objects:

$$\underbrace{\nabla_\theta \ell_\tau(x; \theta)}_{\text{per-example (random)}}, \quad \underbrace{g_\tau(\theta) = \mathbb{E}_{x \sim D_\tau} [\nabla_\theta \ell_\tau(x; \theta)]}_{\text{task gradient (the drift)}}, \quad \underbrace{\hat{g}_\tau^{(B)}}_{\text{batch: drift + noise of covariance } \Sigma_\tau / B}.$$

In the HMM setting the tasks share one next-token loss and differ only in conditioning: D_τ is the distribution of contexts given hidden state τ , so g_τ is the state-conditional mean gradient.

Definition 5 (alignment and drive). For a circuit C with projection P_C (Definition 2), the alignment of two tasks on C and the aggregate drive on C are

$$\text{align}(\tau, \tau' \mid C) = \frac{\langle P_C g_\tau, P_C g_{\tau'} \rangle}{\|P_C g_\tau\| \|P_C g_{\tau'}\|}, \quad \text{drive}(C) = \left\| \sum_\tau w_\tau P_C g_\tau \right\|^2,$$

where w_τ is the fraction of tokens that both carry loss and belong to task τ (in the HMM setting: the fraction of positions whose hidden state is τ). With n aligned consumers the drive grows like n^2 ; with unaligned (randomly oriented) consumers like n , no better than noise; with sign-opposed consumers it can vanish at positive per-task norms. For a single consumer, drive is proportional to its weight, which is how the empirical ‘drive law’ $\tau_{\text{learn}} \propto 1/\varphi$ of Figure 3 follows: the use frequency φ is exactly w_τ .

These three regimes are exactly the ones realized by the experiments below (B3, B1, B2 respectively). Each experiment is summarized by one figure; the text records only the task and how the figure was obtained.

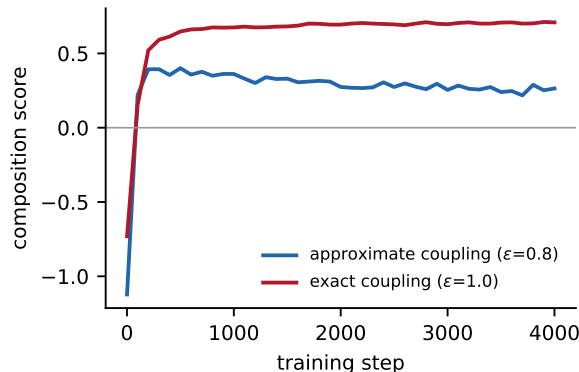


Figure 1: Held-out composition score over training (one run per condition). Approximate coupling ($\epsilon = 0.8$): early peak, then erosion. Exact coupling ($\epsilon = 1.0$): stable and still improving. The solution that replaces the eroding compositional one is more accurate on the *training* masks only; its held-out score is what decays.

B1: compositional solutions are found first, and can erode. Task: next-token prediction on sequences from four independent hidden-Markov factors sharing one output alphabet; each sequence draws a *mask* (one of the $2^4 - 1 = 15$ nonempty subsets of factors), and at each step, with probability ϵ , the active factors emit jointly (copying a designated leader) rather than independently, so the mask is inferable and predictively relevant. A small transformer (4 layers, $d = 128$, 0.8M parameters) trains on 10 masks; the other 5 are held out. The *composition score* on held-out masks is $(\text{CE}_{\text{restricted}} - \text{CE}_{\text{model}}) / (\text{CE}_{\text{restricted}} - \text{CE}_{\text{oracle}})$, where the oracle is the exact Bayes predictor and the restricted reference is the best predictor that does not infer the mask: 1 = full composition, 0 = no better than ignoring composition. Figure 1 plots this score against training step (logged every 100 steps; one run per coupling strength). The compositional solution appears early in both regimes and remains partial (plateau ≈ 0.7 even when stable); under approximate coupling it erodes as an in-distribution-specialized solution takes over (the aligned demand of the training masks decoheres; see section 4).

B2: misaligned gradients are absorbed by splitting. Task: two scalar regression targets computed from a shared latent z , $t_A = \text{relu}(z)$ and $t_B = (1 - \lambda)\text{relu}(z) + \lambda\text{relu}(-z)$, learned by a small ReLU network (8 hidden units) with one shared feature layer, so the dial λ moves the two tasks from identical to opposite-sign use of the same latent. Figure 2 plots three quantities against λ (means over 4 seeds; heterogeneous units share the axis): the realized alignment $\text{align}(A, B \mid C)$ with P_C the restriction to the shared layer’s weights; the normalized error of a network *forced* to route both tasks through a single shared unit; and the Jaccard overlap between the unit sets each task relies on (by per-unit ablation) in an unconstrained network. As alignment crosses to negative, the forced-shared solution degrades while the free network keeps zero error by *reducing* unit sharing ($.73 \rightarrow .41$): misaligned demand is absorbed by partial duplication.

B3: more-used circuits form first, and are chosen. Two measurements. Left of Figure 3: a gated composite task where a subcircuit’s inputs are informative on a fraction φ of examples (the use frequency); we log the step at which the subcircuit’s accuracy on its own targets crosses 0.9, across $\varphi \in \{1.0, 0.3, 0.1, 0.03\}$ (points are 2-seed means). Formation time follows $\tau \propto 1/\varphi$: drive sets speed. Right: a task with two behaviorally equivalent views of the same information, one shared across T tasks and one private; we plot the reliance difference (drop in accuracy when ablating the shared view minus the same for the private view, normalized) against T , alongside the symmetric control in which both views are shared (a second control, forcing the private view, confirms it is fully learnable and is not plotted). The network increasingly routes through the shared view as its reuse grows.

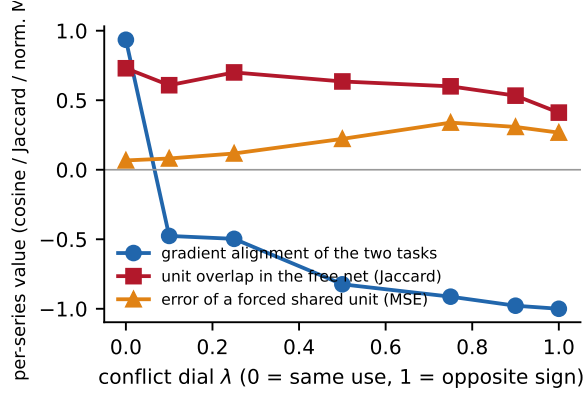


Figure 2: The sign exercise: sweeping the conflict dial λ from coherent to sign-opposed use of a shared feature.

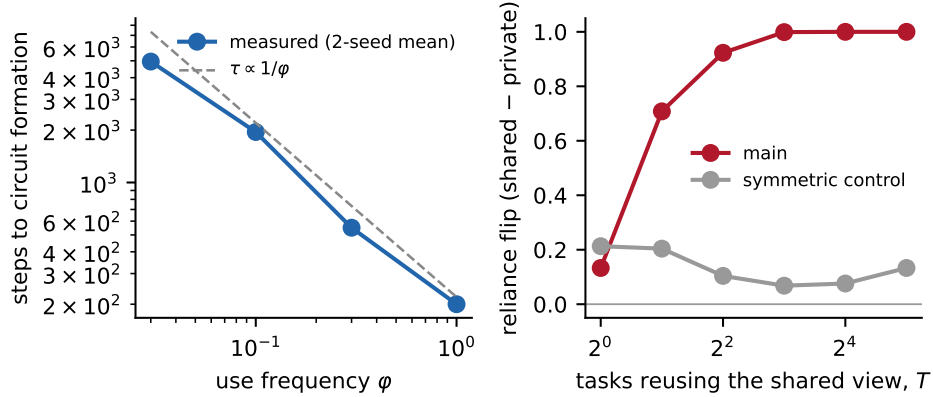


Figure 3: Left: the drive law, formation time vs. use frequency. Right: reuse-scaled selection between equivalent circuits.

Supporting evidence: across ~ 140 networks in two task families, a forward-transfer measure of modularity correlates with behavioral evolvability at matched loss (Spearman 0.65–0.73).

3 Depth: the pressure to consume other units

We now explain a possible incentive for neural networks to learn compositionally deep circuits. At the same time, this also offers an explanation for why networks with a residual stream represent information linearly (explained to me by Francesco). The basic idea is that information that is consumed by many downstream circuits is best represented linearly: the attention and MLP operators add to the residual stream, but with a norm that is relatively small compared to the residual vector in the typical case. Therefore information represented linearly can persist for many layers in a relatively stable format.

The main obstacle for a transformer network is integrating information across many token positions, i.e., computing joint correlation functions in many variables. The difficulty rises with the number of variables because the *drift self-cancels*: for a genuinely joint function of k inputs (a parity is the extreme case), flipping any one input flips the target, so per-example gradients cancel in matched pairs and the population gradient g_τ survives only through higher-order couplings in the weights. The per-example gradients are not small; they point in data-dependent directions that average away. Figure 4 shows this measured directly: we trained a 1-layer transformer on three single-route tasks whose answers are functions of 1, 2, and 3 source

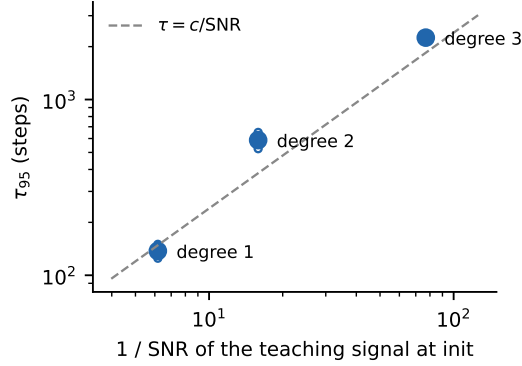


Figure 4: Why higher-order correlations are harder: time-to-learn against the inverse drift-to-noise ratio of the circuit’s teaching signal at initialization, for circuits of input degree 1/2/3 on one architecture. The line is $\tau = c/\text{SNR}$ with a single shared constant.

tokens respectively (a lookup, an xor, a 3-parity; two seeds each), recorded the time τ_{95} to 95% held-out accuracy, and separately measured at initialization the drift-to-noise ratio $\text{SNR} = \|\mathbb{E}\nabla_{\theta}m\|^2 / \mathbb{E}\|\nabla_{\theta}m\|^2$ of the answer margin m (Definition 6; expectations estimated from 2,560 examples at the same initialization for all three tasks). Small open markers in Figure 4 are the individual seeds. Learning time multiplies by ~ 4 per added variable while $\tau \cdot \text{SNR}$ stays constant to within a factor of 2. On tree-structured tasks the attenuation has a closed form: each latent level crossed multiplies the usable squared correlation by the channel’s SDPI constant $\eta_{\ell} = \lambda_2(T_{\ell})^2$, giving a geometric ladder of learning times.

However, when the token stream is such that some tasks are easy and depend on, say, the current token and previous token, and in computing this easy solution the network writes some information to an intermediate layer that can then be combined with a previous token to solve harder tasks, we will see precisely a compositionally deep circuit emerging.

RRXOR is the minimal case of this. The xor function is a two-position circuit, directly financed at every third position, and it forms almost immediately. Once it exists, its *violations* become a new intermediate variable that localizes the phase: the teaching signal for the phase circuit literally does not exist until the xor unit computes. Figure 5 shows the formation: from 79 checkpoints of a single 4-layer next-token run we plot, per position class, the cross-entropy and the probability mass the model assigns to the xor of the previous two bits. The xor circuit forms first and is applied at the right positions only as the phase circuit sharpens.

The same structure iterated at nested scales is the matryoshka cascade (Figure 5, right). The process extends RRXOR to period 27: a super-period of nine consecutive (r_1, r_2, det) blocks in which the deterministic slot obeys $\text{det} = r_1 \oplus r_2 \oplus \text{rule}(b)$, with $\text{rule} = 1$ (XNOR) on blocks $b \in \{2, 5\}$ and $\text{rule} = 0$ elsewhere; block 8 is tracked as its own class (‘flip’) because it is the position where the mod-9 pattern would suggest a third exception but the rule flips back to plain xor. Windows start at a random offset mod 27, so all block identities must be inferred from content. The plot is generated from the training log of one 6-layer run: per class, the CE and the probability assigned to plain xor. The manufactured-gradient signature is the rise of the wrong bet: the model’s probability of answering plain xor on the XNOR class *rises above chance* while that class’s CE spikes, and those errors are what finance the block-classification circuit that then resolves both (colors are per-panel; the two columns use different time axes because the two runs differ $4\times$ in length). A family of tree-structured grammar tasks (studied in the companion repository) generalizes this quantitatively: learning time per level multiplies by the inverse SDPI constant of the level crossed.

Note that the story we are telling here is not necessarily a hard constraint. First, depth-by-bootstrapping is primarily about *speed and staging, never possibility* at our scales: it is a story about sample efficiency, as our experiments illustrate. Second, sharp stage boundaries require *gating* (the later gradient not existing until the earlier circuit forms, as in RRXOR and the matryoshka tower); mere complexity separation gives

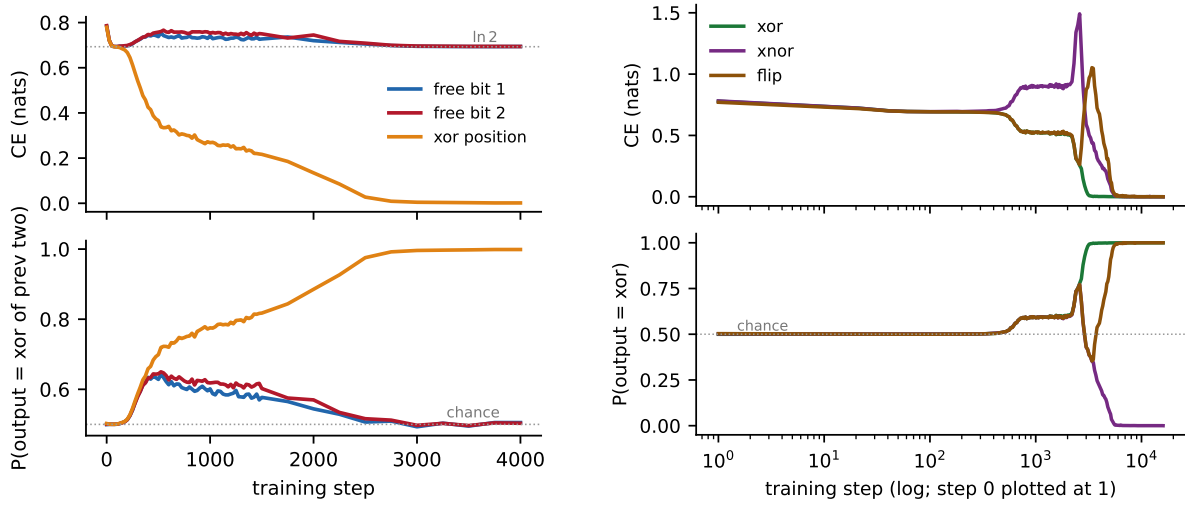


Figure 5: Left: RRXOR formation; per-class CE (top) and probability of outputting the xor of the previous two bits (bottom). Right: the matryoshka cascade (period-27 exception tower over the xor rule: XNOR at two of nine blocks, one block flipping back); the exception classes’ CE *spikes* exactly while the model’s probability of outputting plain xor on them *rises*: the net confidently bets the wrong rule until the correction that distinguishes the blocks is financed by those very errors.

ordered but overlapping onsets.

4 Compositionality is a learning-dynamics notion

The upshot of the previous two sections is that breadth and depth are selected by the same currency: coherent gradient drive. This makes compositionality a learning-dynamics notion rather than an expressivity notion. The expressivity that matters is the set of functions reachable in a short window of SGD from the current parameters; computational units are what make this set large, because aligned consumers accumulate drive on shared units, so new tasks inside the units’ span are learned at composition cost rather than from-scratch cost. This section makes the currency itself measurable, and closes with the sharpest illustrations we have.

4.1 The accessibility of a circuit

Definition 6 (end-to-end accessibility). Decompose the per-example gradient as $\nabla_{\theta}\ell(x) = \bar{g} + \xi(x)$, with $\bar{g} = \mathbb{E}[\nabla\ell]$ the *drift* and ξ the mean-zero *diffusion*. Over T steps drift accumulates coherently ($\sim T$) while diffusion random-walks ($\sim \sqrt{T}$), so a circuit separates from noise after $T^* \sim \text{noise}/\text{drift}^2$. For a circuit f with a decision, let $m_f = \text{logit}(\text{correct}) - \text{logit}(\text{incorrect})$ at f ’s output slots (for decisions with more than two answers, subtract the strongest incorrect logit; the pilot tasks below are binary) and define

$$G_{\text{disc}}(f \mid \theta) = \|\mathbb{E}_x[\nabla_{\theta} m_f]\|^2, \quad \text{SNR}(f) = \frac{G_{\text{disc}}(f \mid \theta)}{\mathbb{E}_x\|\nabla_{\theta} m_f\|^2}.$$

By Jensen’s inequality $\text{SNR} \leq 1$: it is the fraction of the gradient’s power that is systematic. The margin projection is essential. If one computes the same ratio with the raw per-example loss gradient $\nabla_{\theta}\ell$ in place of $\nabla_{\theta} m_f$, the result is dominated by structure shared by every circuit: in the experiment of Figure 4 the raw ratio comes out 0.99 for all three routes alike, because the coherent part of the loss gradient is overwhelmingly the calibration signal (“answers are drawn from the answer alphabet, roughly uniformly”), which is the same for easy and hard routes, and dwarfs the discrimination signal that distinguishes them.

The margin functional cancels any parameter direction that moves both candidate logits together, and keeps exactly the discrimination content.

Figure 4 is the validation: $\tau \propto 1/G_{\text{disc}}$ with a shared constant across circuits of input degree 1–3. Classical difficulty notions (input degree, information exponent, multiplicative chain length) become statements about how G_{disc} scales; the hard zeros of the information-exponent idealization appear as smooth attenuation at finite initialization scale and softmax temperature.

Definition 7 (per-interface accessibility). Fix a *site* k : a layer and a position class, with activation vector $h_k(x) \in \mathbb{R}^d$ at that site on input x , and fix a feature map ϕ (linear, $\phi(h) = h$, unless stated otherwise). For a matrix W , let f_W be the network that is identical to f except that the activation at site k is replaced by

$$h_k(x) \mapsto h_k(x) + W \phi(h_k(x)).$$

At $W = 0$ this is exactly the original network, so the gradient of the training loss with respect to the inserted parameters is well defined at $W = 0$ and equals

$$\nabla_W \ell|_{W=0} = \delta_k \otimes \phi(h_k), \quad \delta_k(x) := \frac{\partial \ell(x)}{\partial h_k},$$

the outer product of the backpropagated error at the site with the features offered there. In words: this is the teaching signal that a new piece of computation inserted at site k would receive on its first step of training (we call the insertion an *adapter*, in the sense of parameter-efficient fine-tuning: a trainable module initialized to have no effect). No adapter is ever actually trained; it is a measurement device. Define

$$G_{\text{disc}}(k | \theta) = \frac{\|\mathbb{E}_x[\delta_k(x) \otimes \phi(h_k(x))]\|^2}{\mathbb{E}_x\|\delta_k \otimes \phi(h_k)\|^2}.$$

The numerator is a cross-covariance between *demand* (δ_k : what downstream coherently asks for at this site) and *supply* ($\phi(h_k)$: what upstream currently offers there); it vanishes in two distinguishable ways, when downstream is unformed (δ_k has no coherent direction) or when upstream is ($\phi(h_k)$ lacks the needed correlate). The projection P_C of Definition 2 specializes to exactly this construction for interface circuits. Sites in our experiments are residual-stream vectors at a given layer and position class. One further variant: when ground truth for a candidate variable y is available from the generative model, replacing $\phi(h_k)$ by $y(x)$ in the numerator measures the demand for that specific content at the site, independent of whether the network can currently supply it; Figure 6 (left) uses this form.

Figure 6 applies this to the matryoshka run: the site is the residual stream after block 4, at deterministic-slot query positions late in the window; across 76 checkpoints we computed the demand for each ground-truth part variable, using the ground-truth form above (per-example δ normalized to unit length, variables standardized; the plotted quantity is the squared norm of their cross-covariance, a squared-correlation-like number in $[0, 1]$), and the supply of u as the held-out R^2 of a ridge probe for u in the stream at the same site. Demand for u is present from step 0 and collapses once its supply forms; demand for the block code is absent at initialization, is *manufactured* as u ’s supply rises, peaks during the exception-stage loss drop, and decays once satisfied. Demand is manufactured by upstream formation, destroyed by satisfaction, and (below) decohered by conflicting consumers.

The same instrument re-reads the B1 erosion (Figure 6, right), on the same $\epsilon = 0.8$ run as Figure 1, at its 21 saved checkpoints. For each training mask c we computed its credit vector $g_c = \mathbb{E}[\partial \ell_c / \partial h_{\text{final}}]$ (mean gradient of that mask’s loss with respect to the final-layer activations), removed the component in the factor-readout subspaces so that what remains is binding-directed, and plot the mean pairwise cosine of the ten masks’ residual credit vectors alongside the held-out composition score. This cosine is the alignment factor inside the drive $\|\sum_c w_c g_c\|^2$: per-mask magnitudes persist throughout, so when the cosines decay the aggregate drive on the shared binding machinery shrinks. The transience of B1 is co-financing decoherence. [single seed; noisy]

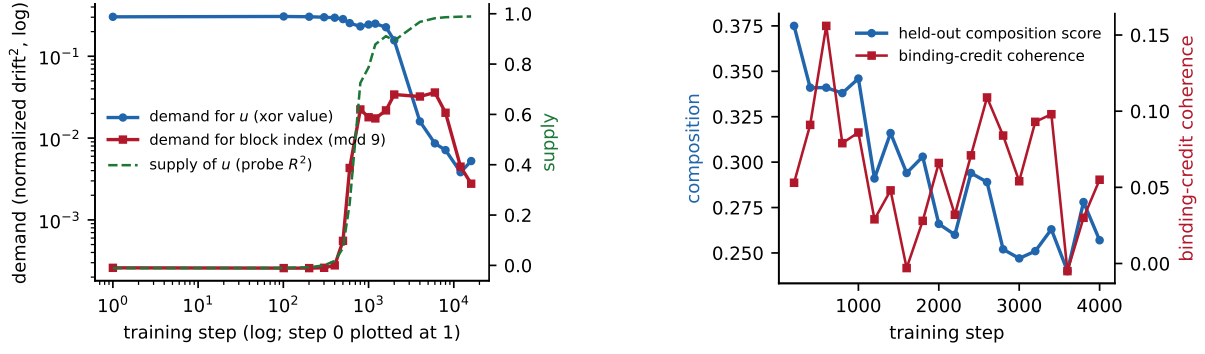


Figure 6: Left: demand and supply at one interface of the matryoshka run. Right: co-financing decoherence in the composition task; held-out composition erodes as the cross-task coherence of the binding-directed credit decays (all 21 checkpoints of the $\epsilon = 0.8$ run; single seed).

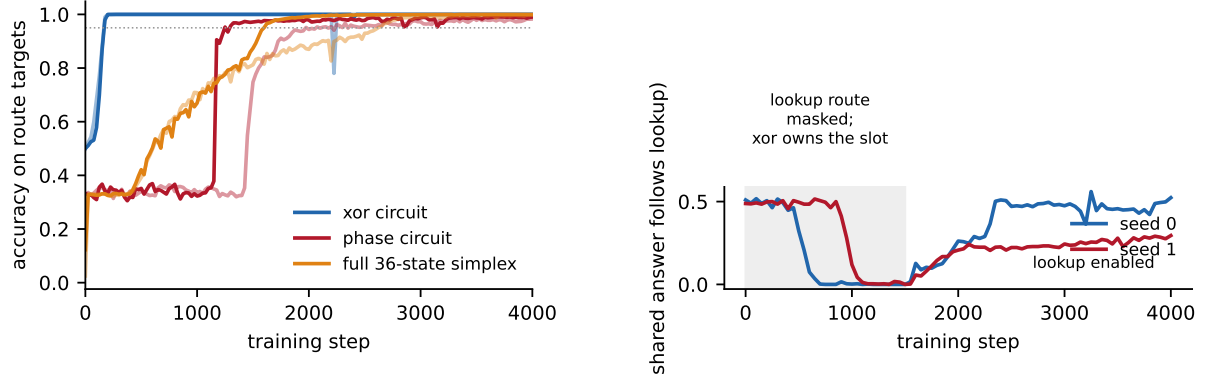


Figure 7: Left: the RRXOR route race under direct supervision (τ_{95} : xor ~ 175 ; phase ~ 1250 – 1975 ; full simplex ~ 1600 – 2625 ; two seeds each). Right: selection is competition, not lock-in: a shared prediction owned outright by the harder (xor) route, trained for 1500 steps with the lookup route masked, is taken over once the lookup route is enabled; preference measured on conflict trials that never appear in training.

4.2 Which solution forms: the RRXOR race

The full 36-state belief simplex and the modular xor-then-phase decomposition are both zero-loss solutions of the same task; the network takes the second path, and we can now offer a quantitative explanation. We raced the routes with best-case direct supervision (Figure 7): three supervised heads on the same 4-layer architecture and data stream, with targets given by (i) the exact 36-way belief-state index from the bisimulation-minimized filter, (ii) the xor of the previous two bits, (iii) the 3-way phase. Even the simplex’s floor price exceeds the modular path’s slowest part, and in the actual next-token task it is worse than slow: the loss only ever demands the decision-relevant quotient of the belief state, so the fiber beyond that quotient has $w \approx 0$ and is never financed (probes confirm it never forms in the deterministic task). Solutions are ordered by accessibility and financing, not by loss. The unsupervised cascade pays the same per-stage prices as these supervised floors, so credit assignment is not the bottleneck for financed parts.

When several adequate circuits compete for one consumer, the arbitration follows a lawful order which we measured with route-competition grids and the crossover of Figure 7 (right): accessibility dominates (a lookup route beats an xor route at preference 1.00 even at $8\times$ weight disadvantage, surviving a complexity-reversal control); weight is a secondary, soft, quickly-saturating bias among accessibility-matched routes; incumbency contributes friction ($\sim 50\times$ slower takeover) but not permanence. Selection is ongoing competition under $G_{\text{disc}}(\theta_t)$: history-dependent but not history-locked.

4.3 Open

(i) Consolidation: re-derive the formation threshold (section 2), the dilution ladder (section 3), and the selection law from one object, $w_k \cdot G_{\text{disc}}(k \mid \theta_t)$; the SDPI products should fall out as its closed form on trees. (ii) Scaling of the per-degree attenuation with width, length, and vocabulary. (iii) Seeds for the decoherence claim. (iv) Path dependence: integrated $\int w G dt$ versus endpoint values as the predictor of selection. (v) Below-threshold learning is diffusion-driven, not absent (hidden progress): where spike-onset variance lives. (vi) Presentation debts of this draft: seed batteries and error bars for Figures 1–3; the B1 construction spelled out; breadth and depth measured rather than used as vocabulary.